

Air Purification — Innovation Recommendation

Versuni Disruptive Innovation | Innovation AI Expert case study | 5 slides All evidence is real: 10,547 real Amazon reviews (237 real, hand-validated purifier products), 2 real market-research sources, 12 real trend documents.

Slide 1 — The recommendation

Pursue: Reliability-Verified Air Purifiers — an extended-life guarantee paired with a real-time self-diagnostic (connected telemetry that flags degrading performance before the unit silently dies).

Two spaces were seriously considered and rejected: **Whisper-Quiet Night Mode** and **Smart/Connected Feature Expansion**. Both are killed on the same three measures — see Slide 4.

Held constant throughout: **Friction Prevalence %**, **CSAT Impact**, **Price-Weighted Exposure** (defined Slide 2).

Slide 2 — Framing (Q1)

- **Category:** Air Purification (Smart Home) — chosen for its clean, real review signal on hardware/software frictions.
- **Consumer group:** Real owners of residential air purifiers who left a review on Amazon.com, 2004–2023.
- **Three questions:** (1) What actually breaks the purifier ownership experience? (2) What would owners pay to fix it? (3) Which opportunity is worth building next?
- **Three measures, fixed and unchanged from here on:** 1. **Friction Prevalence %** — share of real reviews carrying the friction. 2. **CSAT Impact** — mean star rating of affected reviews minus corpus mean. 3. **Price-Weighted Exposure** — sum of real observed prices across affected reviews (a calculated diagnostic, not a survey) — explicitly **not** a revenue, market-size, or willingness-to-pay estimate; see Slide 3.

Slide 3 — What the real data shows (Q2–Q4)

10,547 real reviews, 237 real hand-validated air purifier products, assembled from the McAuley-Lab Amazon-Reviews-2023 public dataset. Product inclusion required two rounds of classifier tightening after manual inspection caught real false positives (replacement filters, vacuum cleaners, wearable necklaces, ozone generators) — see [data_quality_report.md](#).

Defects found (not planted): 239 rating/text sentiment conflicts (2.3% of reviews — using a negation-aware detector after an early version misread "I cannot recommend" as *positive*, see [ai_use_log.md](#)); 18 empty-text rows; **zero** duplicate-text bursts and **zero** volume anomalies — a real, honest finding: this corpus does not contain the kind of promotional review-bombing a synthetic fixture would plant by design.

Friction taxonomy, induced bottom-up from real 1–2★ review text (corpus mean 4.172★, n=10,547):

Theme	Prevalence %	CSAT Impact	n reviews
Noise / motor sound	3.50%	−0.571★	369
Ozone / smell / irritation	0.81%	−1.102★	85
Reliability / stopped working	1.58%	−2.223★	166
Perceived value / effectiveness	0.77%	−2.724★	81
Customer service / returns	0.34%	−2.429★	36
Filter cost / replacement	0.26%	−1.252★	27

92.74% of reviews match none of these six themes — most real reviews are either straightforwardly positive or describe frictions this taxonomy doesn't capture, stated honestly rather than forced into a bucket.

Willingness to pay (Q4): no direct or proxy WTP measurement exists in the assembled real evidence. No conjoint/Gabor-Granger data, and no real consumable-switching behaviour is obtainable from a public review export. What's real: **75 of 237 products have an observed real price** (median \$79.99, range \$3.81–\$1,100), used only as a *price-weighted exposure* indicator (reliability: \$26,357.86; noise: \$84,015.62) — explicitly a relative reach×price signal, not a dollar forecast.

Slide 4 — Why Reliability wins, and what killed the other two

Opportunity space	Friction Prevalence %	CSAT Impact	Price-Weighted Exposure
Reliability-Verified Purifiers	1.58%	−2.223★	\$26,357.86
Whisper-Quiet Night Mode	3.50%	−0.571★	\$84,015.62
Smart/Connected Feature Expansion	1.02%	n/a (mostly praise)	n/a

This is a genuine **Pareto trade-off, not a dominant winner** — stated plainly rather than manufactured into a formula. Noise reaches more reviews and touches higher-price products, but is the *shallowest* satisfaction hit of any real theme. Reliability reaches fewer reviews but the hit is severe — close to the worst in the whole taxonomy — and a "stopped working" failure plausibly ends the customer relationship outright (an inference from the review text, which frequently pairs failure with return/refund language — not a measured churn or repurchase figure, which this evidence cannot provide), unlike noise, which annoys a buyer who usually keeps the unit.

Killed — Whisper-Quiet Night Mode: CSAT Impact of −0.571★ is the *shallowest* satisfaction hit of any real theme, despite the highest prevalence. Mentioned often; fixing it moves satisfaction the least.

Killed — Smart/Connected Feature Expansion: 1.02% real prevalence (107 of 10,547 reviews) — the smallest of the three candidates — and on manual inspection, most of those mentions are **feature praise, not complaints** ("It is incredibly powerful and [connectivity feature]... impressed with it"). No real friction signal to build a roadmap bet on.

Slide 5 — Prize, first test, and what would change our mind

- **Directional signal:** reliability failures ("stopped working," "never worked," "died") appear across brands and across the full 2004–2023 span of this real corpus — a complaint pattern that persists across brands and years *within this review sample*, not a one-brand defect. (This corpus cannot establish what the industry has or hasn't solved outside it.)
- **Most sensitive assumption, stated plainly:** this recommendation rests on a judgment call — that a *severe-but-narrower* friction (reliability) is worth prioritizing over a *shallow-but-broader* one (noise). The opposite judgment — reach matters more than depth, because a mildly annoyed majority still drives more aggregate churn — is equally defensible and would flip the recommendation to Whisper-Quiet Night Mode. The data does not resolve this alone.
- **First experiment:** cross-reference real failure-mention reviews against each product's real rating_number (a popularity proxy) to see whether failures cluster in specific brands/price tiers (→ a competitive positioning bet) or spread evenly across the category (→ a category-wide manufacturing/QA opportunity).
- **Abandon signal:** if failures concentrate in 2–3 specific older/discontinued products rather than spreading across the category, treat this as solved-or-solving and abandon.
- **Q5 sensitivity, actually re-run, not asserted:** re-running `src/real/decision_framework_real.py --market-scenario=imarc` (6.54% CAGR vs. Mordor Intelligence's 5.37%) produces an **identical verdict** — the Price-Weighted Exposure here is built from review-level price exposure, not category CAGR, so the Q5 disagreement changes the category-sizing narrative but not which opportunity wins.